COM115 Week 2 Lab

Solving Equations and Plotting Graphs

Quadratic Equations

As discussed in section 1.9 of the lecture notes, quadratic equations can be expressed in the form

$$ax^2 + bx + c = 0$$

where a, b and c are real numbers. The solutions, i.e. the values of x that solve the equation are obtained from the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

In other words there are two solutions

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

Example 1.10.1 in the notes shows how to work out the solutions for the equation

$$2x^2 + 7x - 4 = 0$$

Since $a = 2$, $b = 7$ and $c = -4$ in this case, these values can be put into the formula to generate the solutions $x = -4$ and $x = 0.5$ (see example 1.10.1 for details).

The following code shows how to calculate the answers in Python. (Note: this code would not work if $b^2 < 4ac$ since it would involve taking the square root of a negative number which we won't consider here.)

In [1]:

```
import numpy as np
a = 2; b = 7; c = -4
x1=x2=0
# First solution
x1 = (-b + np.sqrt(b**2 -4*a*c))/(2*a)
print(x1)

# Second solution
x2 = (-b - np.sqrt(b**2 -4*a*c))/(2*a)
print(x2)
```

```
0.5
-4.0
```

Exercise 1

Use the code above to write a Python function that calculates the two solutions for a quadratic equation of the form $ax^2 + bx + c = 0$. You should then call the function to check that it gives the right answers for the example above. You should also call the function to solve the quadratic

equation in Exercise 1.7.1 in the notes, i.e. $x^2 - 4x - 21 = 0$.

In [ ]:

```

```

Plotting Data Using Matplotlib

Matplotlib (https://matplotlib.org/) is a very useful library for producing figures in Python. (We will also come across another plotting library called *Seaborn* which is based on Matplotlib.) Here we will use the *Pyplot* interface within Matplotlib. For further details, see https://matplotlib.org/stable/tutorials/pyplot.html (https://matplotlib.org/stable/tutorials/pyplot.html) You should look at these resources to explore more details about how to plot and edit figures.

Here is a simple example to illustrate how to produce a x-y line plot.

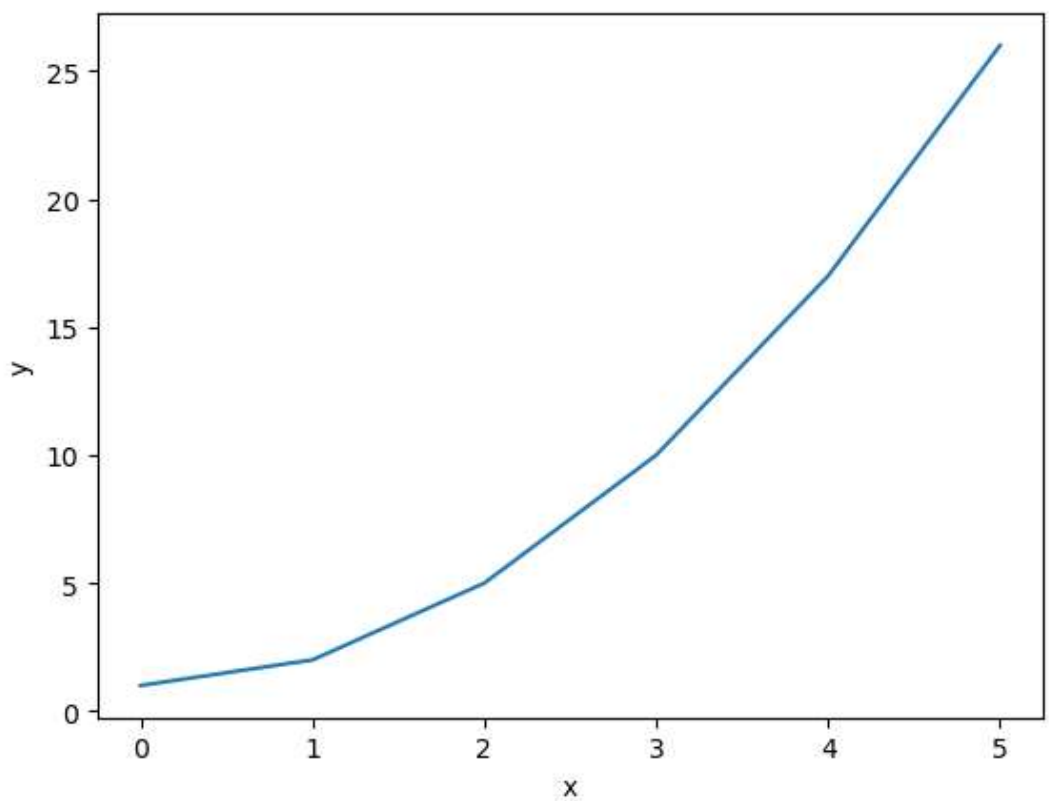
In [2]:

```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([0,1,2,3,4,5])
y = np.array([1,2,5,10,17,26])

plt.plot(x, y)
plt.xlabel('x')
plt.ylabel('y')

plt.show()
```



Exercise 2

Recall the the coursework and exam marks for a group of students (listed in the same order in each case) from Exercise 4 in last week's lab.

- Coursework: 54, 67, 40, 82, 72, 43, 33, 73, 53, 80, 42, 63, 88, 29, 55, 60.
- Exam: 59, 61, 37, 78, 67, 36, 40, 68, 48, 59, 36, 60, 91, 25, 54, 64.

Produce a scatter plot for the coursework vs exam results. How would you describe the relationship between coursework and exam marks? What would be a reasonable exam mark prediction for a student who got 50% in coursework? We will return to this topic when we are covering statistics.

In [ ]:

```

```

Plotting Functions Using Matplotlib

To plot a mathematical function we can use the Numpy function `linspace` to create an array of x values within a specified range and then use the equation to determine the values of another array y. The example below plots the function $y = 3x - 2$ for x values from 0 to 5. Here 100 x values are used. (Note: a legend is included to label the plot. This becomes more relevant when there are multiple plots. See exercise 3.)

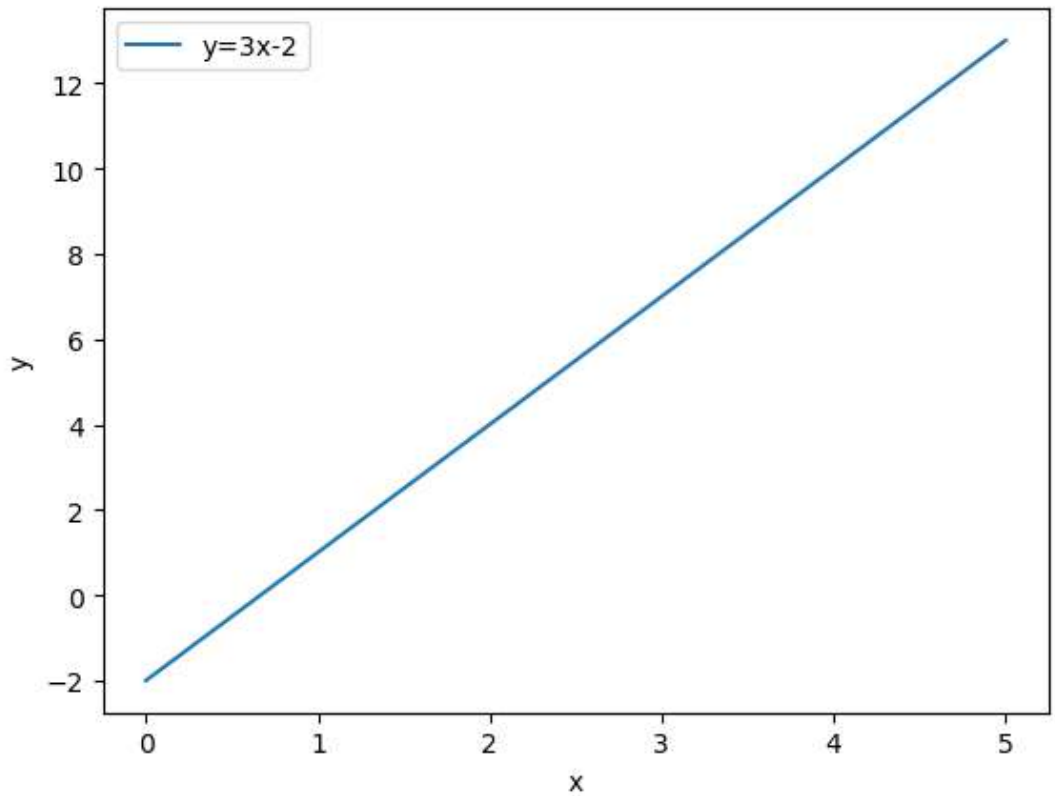
In [3]:

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0,5,100)
y = 3 * x - 2
plt.plot(x, y, label='y=3x-2')
plt.xlabel('x')
plt.ylabel('y')

plt.legend()

plt.show()
```



Exercise 3

Straight lines can be expressed in the form $y = mx + c$, where m is the gradient and c is the intercept. In the example above $m = 3$ and $c = -2$. In the cell below try varying the values of m and c to determine what effect they have? Try including multiple plots in the figure by using multiple calls to `plt.plot` . What effect do large or small or negative values of m and c have?

In [ ]:

```

```

Exercise 4

Plot the following function for values of x between -5 and 5

$y = 2x^2 + 7x - 4$

How would you plot a dashed black line instead of a solid blue line?

In [ ]:

Exercise 5

In Example 1.10 in the lecture notes, we obtained solutions for the equation $2x^2 + 7x - 4 = 0$, i.e. $x = -4$ and $x = 0.5$. How do these solutions relate to the plot of $y = 2x^2 + 7x - 4$? Basically, we need to find the values of x where this curve crosses the x-axis, i.e. the values of x where $y = 0$. It might useful to include gridlines in the plot of $y = 2x^2 + 7x - 4$ using the command `plt.grid(True, which='both')` . You can then check that the points where the curve intersects $y = 0$ are found at $x = -4$ and $x = 0.5$. Try this out below.

In [ ]:

Exercise 6

Can you plot a function to show why the equation $x^2 - 8x + 16 = 0$ in Example 1.10.2 has only one solution?

In [ ]:

Exercise 7

The cell below plots the $cos(x)$ function (see section 1.12 of the lecture notes). Use further cells below to plot $cos(2x)$, $cos(0.5x)$ and $cos(x - 90)$. Which other function is the same as $cos(x - 90)$? Look at the other functions in section 1.12 of the lecture notes.

In [1]:

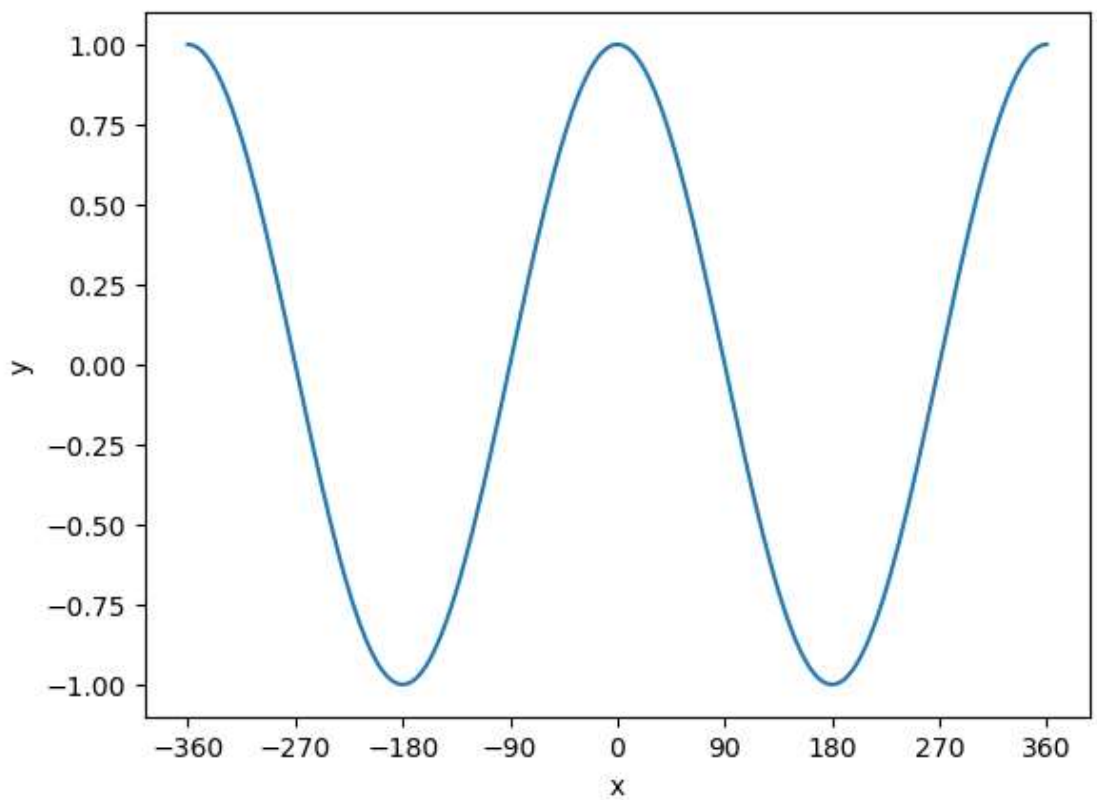
```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-360,360,200)
y = np.cos(np.radians(x)) #The cos function using radians so we need to conver

plt.plot(x, y)

x_grid_points = [-360, -270, -180, -90, 0, 90, 180, 270, 360]
plt.gca().xaxis.set_ticks(x_grid_points)
plt.xlabel('x')
plt.ylabel('y')

plt.show()
```



In [ ]:

In [ ]: